CableWhale AI Ltd Network

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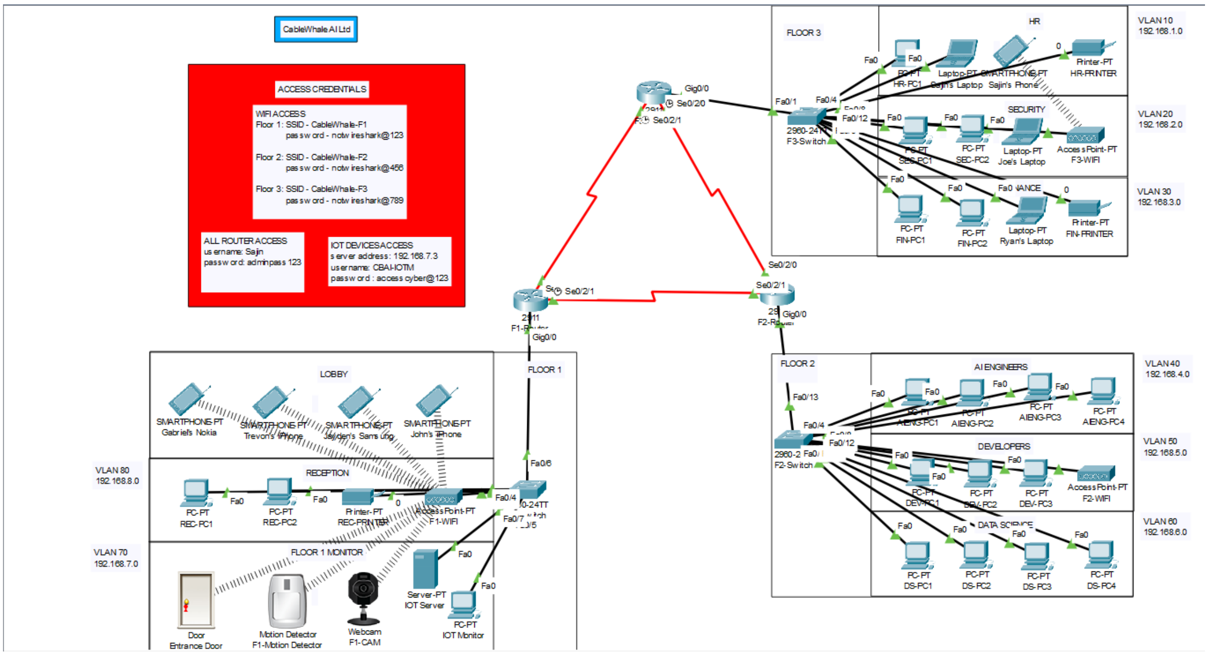
Overview

This project is made to simulate a basic network topology of a small company that develops its own AI model. This network is made for one of their buildings; which consists of 3 floors, each with their own departments.



Configurations of Each Department

Department	VLAN	Subnet	Devices
HR	10	192.168.1.0/24	1x PC 1x Laptop 1x Smartphone 1x Printer
Security	20	192.168.2.0/24	2x PC 1x Laptop
Finance	30	192.168.3.0/24	2x PC 1x Laptop 1x Printer

AI Engineers	40	192.168.4.0/24	4x PC
Developers	50	192.168.5.0/24	3x PC
Data Science	60	192.168.6.0/24	4x PC
Floor 1 Monitor	70	192.168.7.0/24	1x Server 1x PC
Reception	80	192.168.8.0/24	2x PC 1x Printer

OSPF

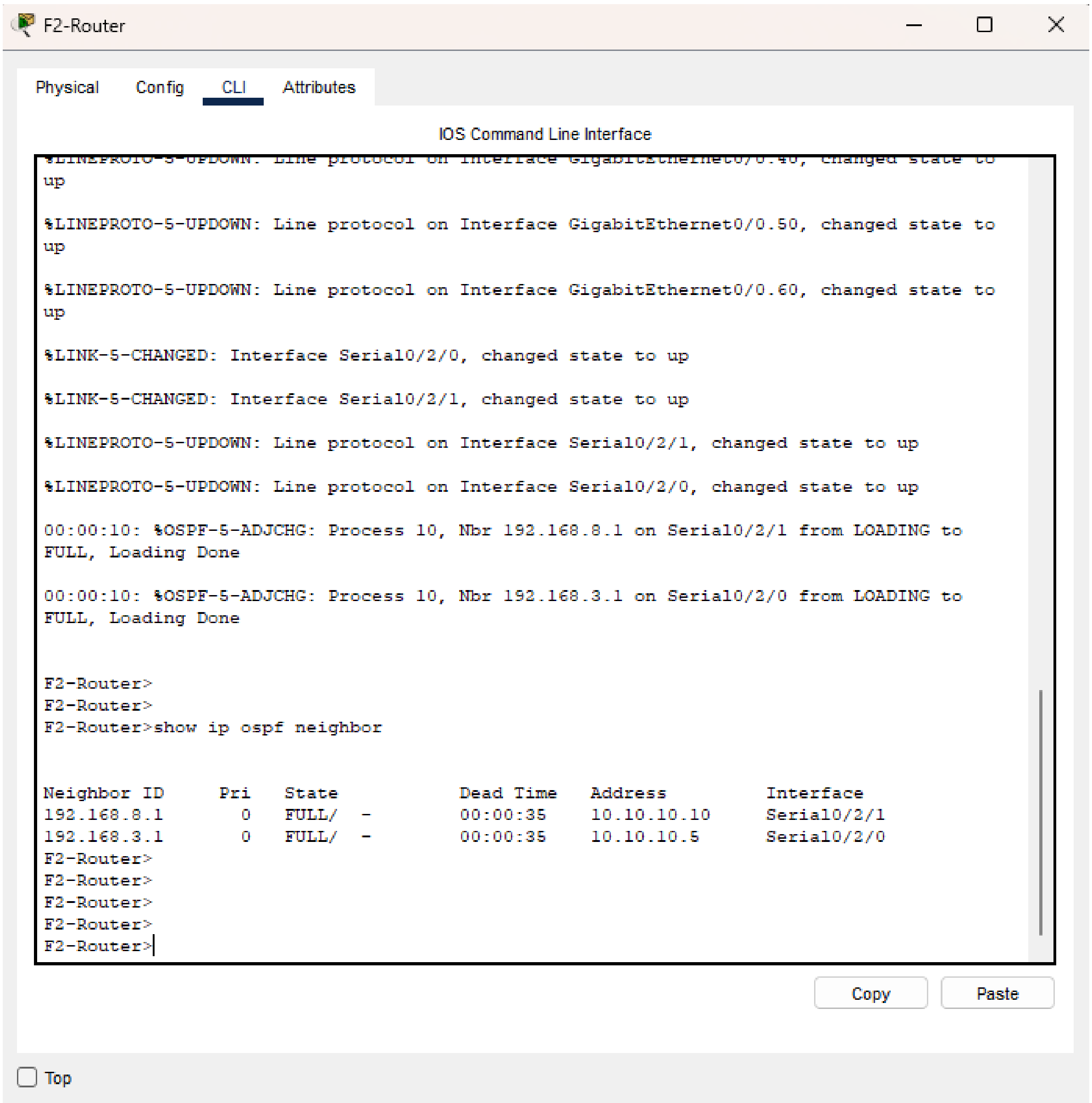
All routers are configured with OSPF, all in a single area (area 0 in this case).

CableWhale AI Ltd's network operates across 3 floors with each having their own departments. This means that using OSPF allows for a faster and more efficient way of accessing shared resources and communication between devices.

Benefits

- Reduces the need for manual configuration when a new floor or department is added.
- Overall improved communication speed.

Validation



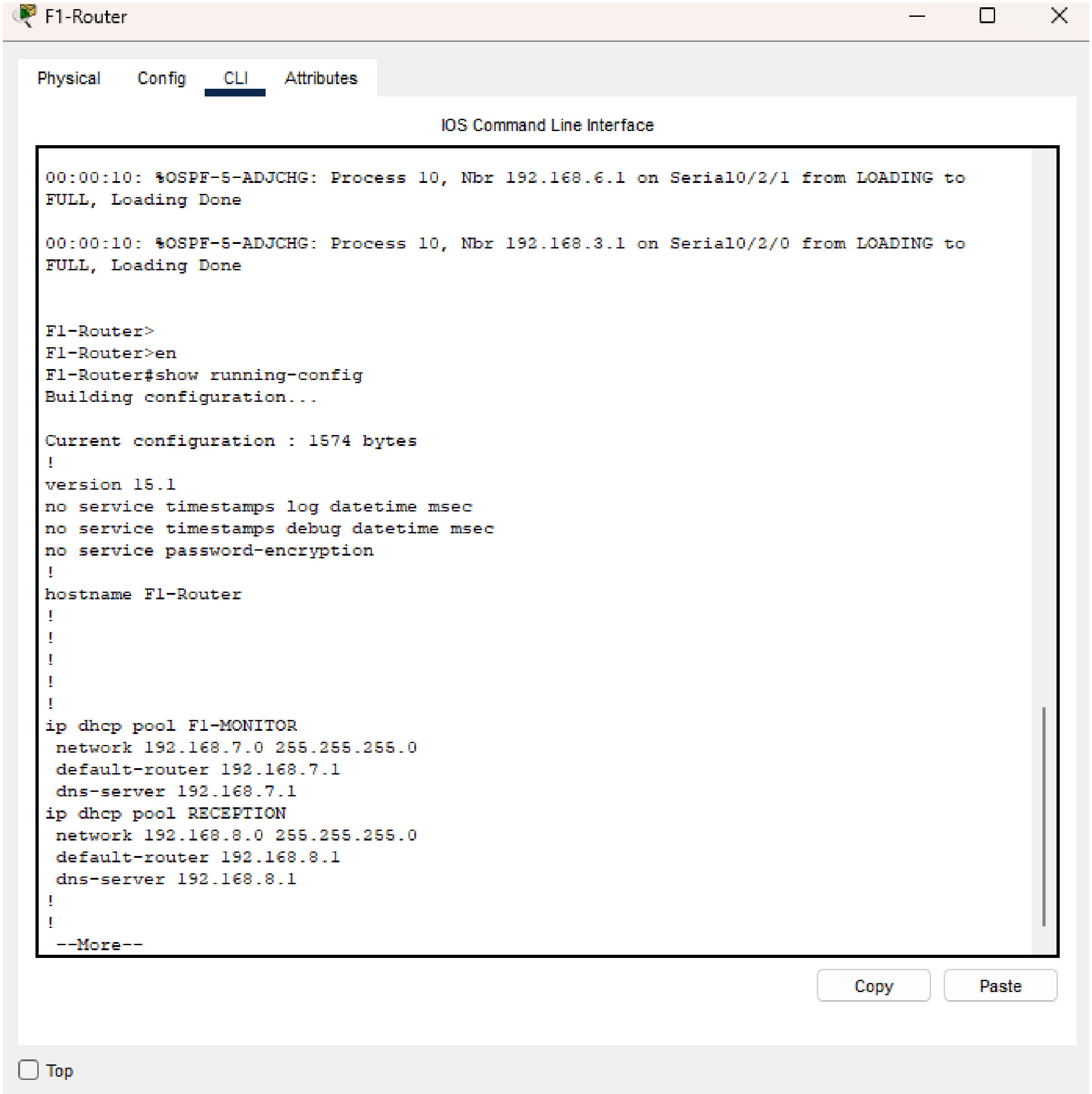
DHCP

Since the company has multiple devices such as laptops, PC's and smartphones; using DHCP will easily increase efficiency in the network.

Benefits

- Reduces errors related to manual configuration.
- Allows faster scalability as adding a new device to the network is easy.
- Saves the overall time required by the IT team to set up the network.

Validation




```
00:00:10: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.6.1 on Serial0/2/1 from LOADING to
FULL, Loading Done

00:00:10: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.3.1 on Serial0/2/0 from LOADING to
FULL, Loading Done


F1-Router>
F1-Router>en
F1-Router#show running-config
Building configuration...

Current configuration : 1574 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname F1-Router
!
!
!
!
!
ip dhcp pool F1-MONITOR
 network 192.168.7.0 255.255.255.0
 default-router 192.168.7.1
 dns-server 192.168.7.1
ip dhcp pool RECEPTION
 network 192.168.8.0 255.255.255.0
 default-router 192.168.8.1
 dns-server 192.168.8.1
!
!
 --More--
```

F2-Router

Physical Config CLI Attributes

IOS Command Line Interface

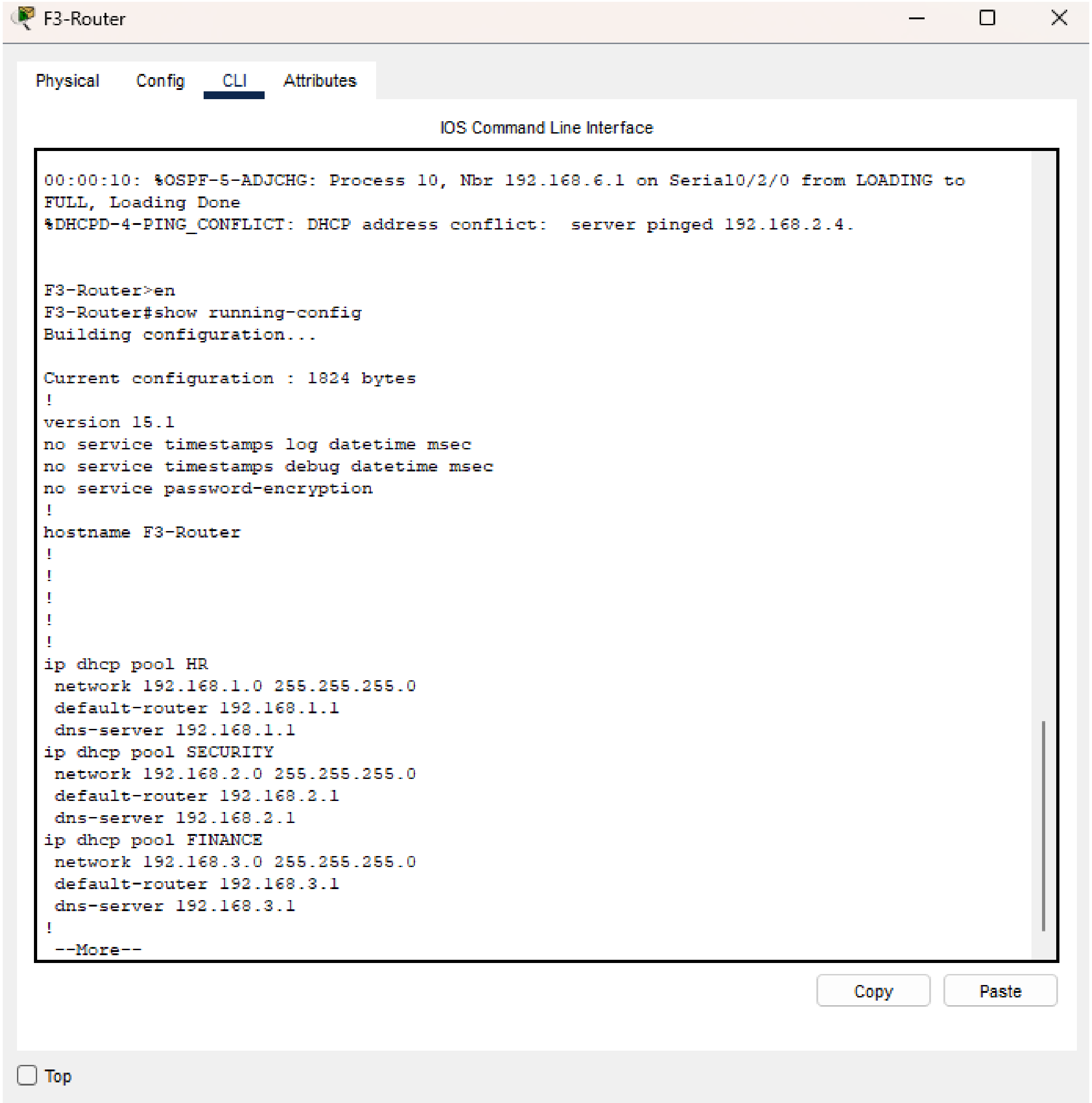
```
00:00:10: %OSPF-5-ADJCHG: Process 10, Nbr 192.168.3.1 on Serial0/2/0 from LOADING to
FULL, Loading Done


F2-Router>
F2-Router>en
F2-Router#show running-config
Building configuration...

Current configuration : 1805 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname F2-Router
!
!
!
!
!
ip dhcp pool AI-ENGINEERS
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
 dns-server 192.168.4.1
ip dhcp pool DEVELOPERS
 network 192.168.5.0 255.255.255.0
 default-router 192.168.5.1
 dns-server 192.168.5.1
ip dhcp pool DATA-SCIENCE
 network 192.168.6.0 255.255.255.0
 default-router 192.168.6.1
 dns-server 192.168.6.1
!
 --More--
```

Copy Paste

Top



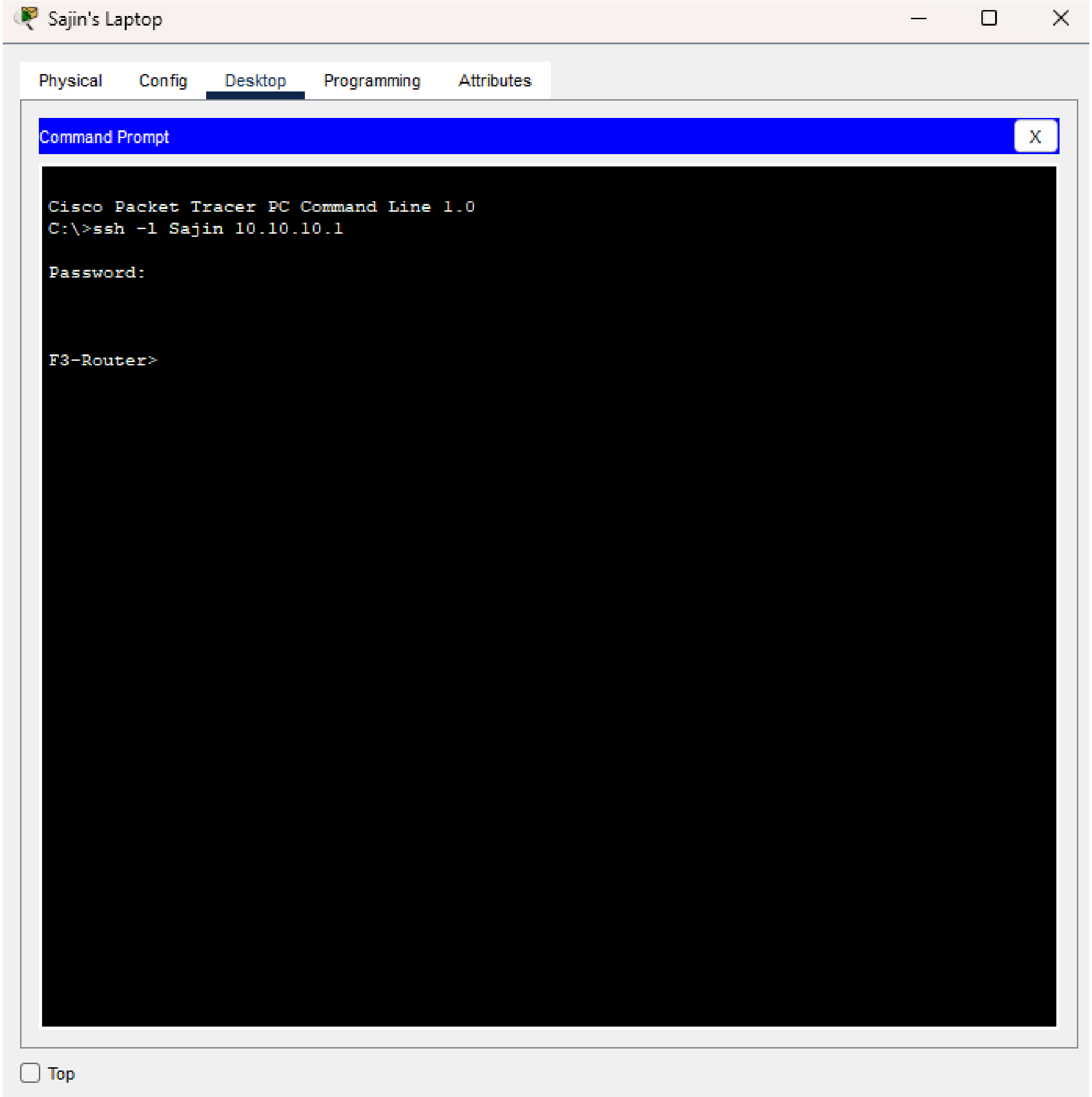
SSH

Developing an AI model means that the routers and switches would need regular maintenance and configuration to a standard. Using SSH allows remote configuration without having the need to physically connect to the devices.

Benefits

- Allows for easy remote configuration.
- Improved security.
- Faster troubleshooting.

Validation



Inter-VLAN Routing

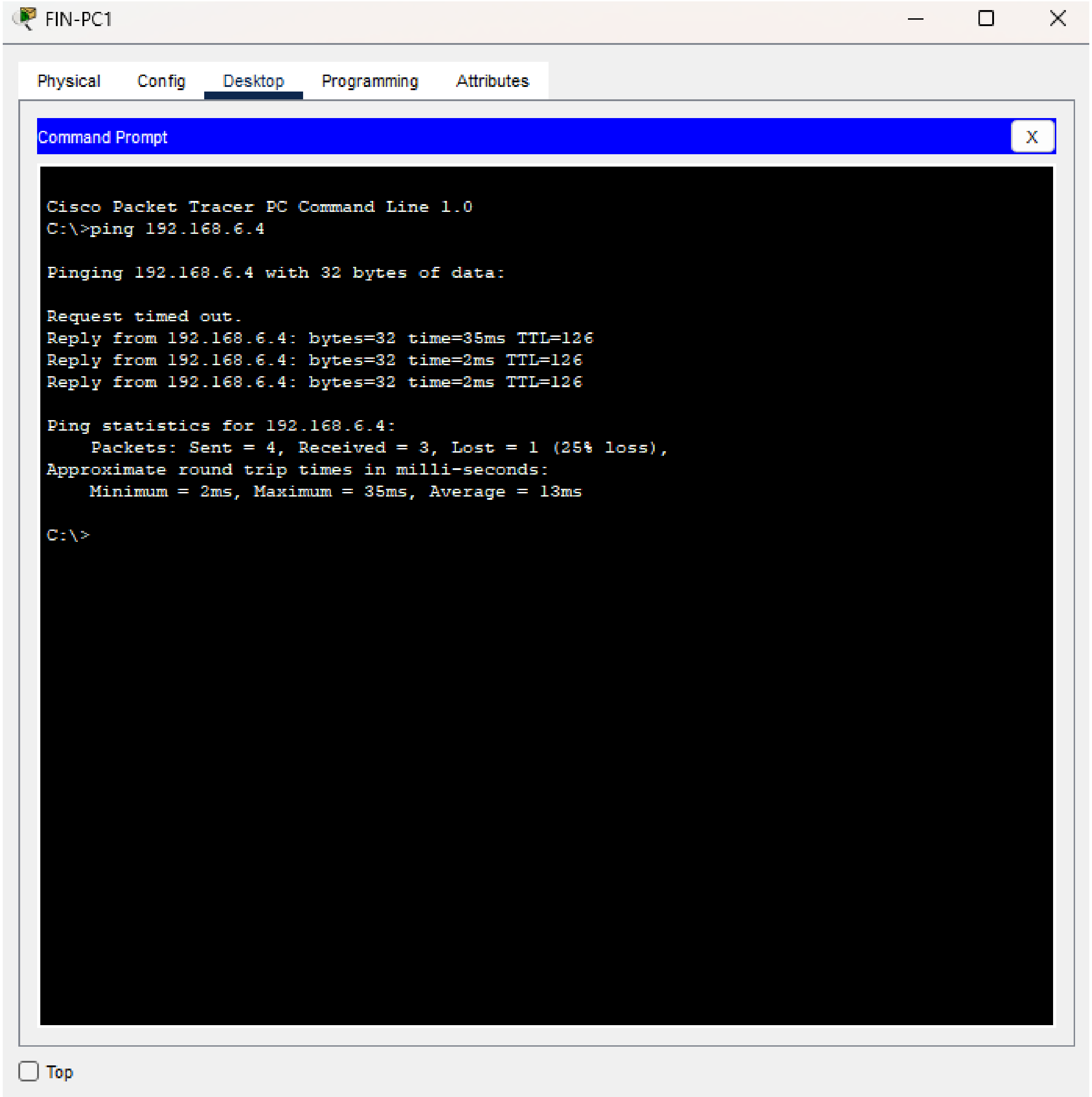
Each department is configured with its own VLAN for proper organization and improved security. However, by default; devices in different VLAN's cannot communicate with each other. Therefore, inter–VLAN Routing allows communication between devices across different VLAN's.

Benefits

- Allows sharing of resources between different departments.
- Collaboration is improved, increasing overall business efficiency.

Validation

Pinging a PC in the Data Science department in floor 2 from a PC in the Finance department in floor 3.



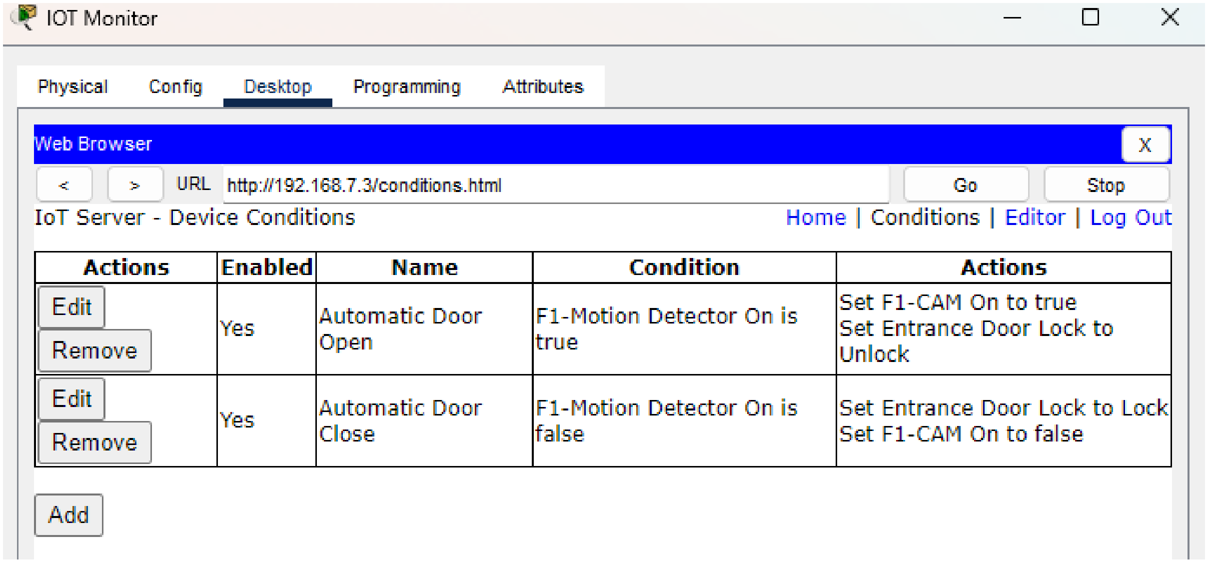
IOT Conditions

The main door of the building uses an automatic door that is also monitored by a CCTV camera.

Conditions

1. The automatic door opens when motion is detected. The CCTV camera then turns on to monitor the motion.

2. After no motion is detected, the door closes and the camera turns off.



Actions	Enabled	Name	Condition	Actions
Edit Remove	Yes	Automatic Door Open	F1-Motion Detector On is true	Set F1-CAM On to true Set Entrance Door Lock to Unlock
Edit Remove	Yes	Automatic Door Close	F1-Motion Detector On is false	Set Entrance Door Lock to Lock Set F1-CAM On to false

Validation

YT vid